

Exp 11:

Write the python program for Tic Tac Toe game

Aim

To write and implement a Python program for the Tic Tac Toe game, where two players take turns placing X and O on a 3×3 board, and the program checks for a winner or a draw.

Algorithm

1. Start the program.
2. Create an empty 3×3 Tic Tac Toe board.
3. Assign Player 1 as X and Player 2 as O.
4. Display the board.
5. Ask the current player to enter the row and column position.
6. Check whether the selected position is empty.
7. If empty, place the player's symbol.
8. Otherwise, ask the player to enter another position.
9. Check if the current player has won by completing:
 10. Any row,
 11. Any column, or
 12. Any diagonal.
13. If a player wins, display the winner and stop the game.
14. If all 9 cells are filled and no player wins, declare the game as a Draw.
15. Otherwise, switch to the other player and repeat from Step 4.
16. End the program.

Program:

```
board = [' ' for _ in range(9)]  
  
# Display the board  
def display():  
    print()  
    print(board[0], "|", board[1], "|", board[2])  
    print("--+---+--")  
    print(board[3], "|", board[4], "|", board[5])  
    print("--+---+--")  
    print(board[6], "|", board[7], "|", board[8])  
    print()
```

```
# Check winner
```

```
def check_winner(player):
```

```
    win = [(0,1,2), (3,4,5), (6,7,8),  
          (0,3,6), (1,4,7), (2,5,8),  
          (0,4,8), (2,4,6)]
```

```
    for x, y, z in win:
```

```
        if board[x] == board[y] == board[z] == player:
```

```
            return True
```

```
    return False
```

```
# Main game
```

```
player = "X"
```

```
for i in range(9):
```

```
    display()
```

```
    pos = int(input("Player " + player + ", Enter position (1-9): ")) - 1
```

```
    if board[pos] == " ":
```

```
        board[pos] = player
```

```
    else:
```

```
        print("Position already occupied!")
```

```
        continue
```

```
if check_winner(player):
```

```
    display()
```

```
    print("Player", player, "Wins!")
```

```
    break
```

```

    if player == "X":
        player = "O"
    else:
        player = "X"
else:
    display()
    print("Game Draw!")

```

Output:

```

Player X, Enter position (1-9): 5

X |   |
---+---+---
O | X |
---+---+---
   |   |

Player O, Enter position (1-9): 6

X |   |
---+---+---
O | X | O
---+---+---
   |   |

Player X, Enter position (1-9): 9

X |   |
---+---+---
O | X | O
---+---+---
   | X

Player X Wins!
>>> |

```

Result:

The Python program for the Tic Tac Toe game was successfully implemented and executed. The program allowed two players to play alternately, correctly detected the winner or a draw, and displayed the final game result successfully.